

Problem Set 1

Due Friday, January 30th, before 11:59pm

Instructions: This is a group assignment. The size of your group must be less than or equal to 3 students. The memberships of groups may change for different homework assignments. Some assignments will be individual assignments.

I suggest that you submit your computations in the form of an Excel or Python file via Canvas. If you upload a Python file, it MUST be a .ipynb file and make sure that you have clearly marked your solutions to each question within the notebook. You may also attach a word or pdf document to answer questions 2 to 4.

(a) Only submit one assignment per group **but make it very clear who the group members are.**

(c) Please make sure that your file is well-organized and clear, with appropriate text, explanations, and formatting. If I cannot figure out what you did, then what you did is wrong.

1. [5 points] I have downloaded some old(ish)¹ market option prices, these are $V_{\text{market}}(K_i)$ values for a range of i values from Jan 9, 2023 from OptionMetrics via wrds (<https://wrds-www.wharton.upenn.edu/pages/get-data/optionmetrics>) I have

- Removed all in-the-money options (they are less traded)
- Removed any option with volume < 10
- Removed any option with a bid/ask = 0

I have chosen an expiry date (T) of March 17, 2023. I have also downloaded relevant continuously compounded risk-free rate (r) data, annualized continuous dividend yield and index data for that date also. Data is presented on Canvas.

You have some tasks to do ☺

- By assuming the Black-Scholes formula and using a volatility of 10% for each option, calculate the Black Scholes value of the options $V(K_i|S_0, \sigma_i = 0.1, d, r, T)$ (be careful as there are calls or puts)
- By varying the volatility, σ_i , of *each option* calculate the implied of each option $V_{\text{market}}(K_i)$. Plot a graph of moneyness ($K_i/S_0 \times 100$) on the x-axis against implied volatility, σ_i , on the y-axis.
- Finally, find the single implied volatility, σ_M , that **minimizes** the sum of the squared error

$$\sum_i [V(K_i|S_0, \sigma_M, d, r, T) - V_{\text{market}}(K_i)]^2$$

of each option. What is σ_M ? Include its value as a horizontal line on your graph from part b.

¹ Optionmetrics data currently is not regularly updated ☺

OPTIONAL TASK (If you are bored)

One easy improvement on our lognormal model could be a mixture of lognormals. Where call² option prices $c(K)$ are given by

$$c(K) = e^{-rT} E^Q[\max(S_T - K, 0)]$$

$$c(K) = e^{-rT} \int_K^\infty (x - K) f_Q(x) dx$$

Where $f_Q(x)$ is typically a lognormal but could be any density function. If a single lognormal has the density $y(x|F, \sigma, T)$ then density is unimodal (has one mode) and has mean equal to the forward price $F = S_0 \exp((r-d)T)$ it's values depend only upon F and $\sigma\sqrt{T}$. A mixture of lognormal distributions is defined by

$$f_Q(x) = p\psi(x|F_1, \sigma_1, T) + (1-p)\psi(x|F_2, \sigma_2, T) \quad x > 0$$

The distribution is assumed to be risk-neutral, so $F = pF_1 + (1-p)F_2$. This adds a constraint to our problem as $F_2 = \frac{F - pF_1}{1-p}$. This gives four parameters to be estimated, $\theta = (p, F_1, \sigma_1, \sigma_2)$. The values of calls and puts are given by:

$$C(K|\theta, r, T) = pC_{BS}(F_1, T, K, r, \sigma_1) + (1-p)C_{BS}(F_2, T, K, r, \sigma_2)$$

$$P(K|\theta, r, T) = pP_{BS}(F_1, T, X, r, \sigma_1) + (1-p)P_{BS}(F_2, T, X, r, \sigma_2)$$

Where here

$$C_{BS}(F, T, K, r, d, \sigma) = e^{-rT} (FN(d_1) - KN(d_2))$$

$$P_{BS}(F, T, K, r, d, \sigma) = e^{-rT} (KN(-d_2) - FN(-d_1))$$

and

$$d_1 = \frac{\ln(F/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, d_2 = d_1 - \sigma\sqrt{T}$$

Task:

- d. Finally, find the parameter set, θ , that minimizes the sum of the absolute % error

$$\sum_i [V(K_i|\theta, r, T) - V_{market}(K_i)]^2$$

of each option. What are the values of the parameters θ ?

² The put option formula can easily be figured out!

- e. From your calibrated option prices $V(K_i|\theta, r, T)$ back out the Black-scholes implied volatilities, σ_i , as in part b above only now for your calibrated $V(K_i|\theta, r, T)$ (not $V_{market}(K_i)$). Add a plot of these σ_i against moneyness to your graph from part c. above. How well does the mixture of lognormals do at replicating the market implied volatilities??
2. **[2 points]** You are interviewing with a trading firm in Chicago that trades equity options listed on the CBOE.
- a. You are given the following hypothetical scenario:
- “You have a portfolio of options on DEF stock, which is trading at \$55. Your Δ is 8,000, and your Γ is $-4,000$. Suddenly the stock price drops to \$50. How much money did you just lose?”
- How should you answer? (If you answer incorrectly, the interview is over, and you will have to find a marketing job.)
- b. Please sketch a graph of the value of the portfolio as a function of the price of the underlying stock.
3. **[2 points]** You are interviewing with another trading firm in Chicago that trades equity options listed on the CBOE. You are given the following hypothetical scenario:
- “You have a portfolio of options on DEF stock, which is trading at \$55. Your Δ is 8,000, your Γ is $-4,000$, and your vega is $-5,000$. (In interpreting this vega, a change from $\sigma = 20\%$ to $\sigma = 21\%$ is one unit, not 0.01 unit.) The options in your portfolio have been trading at an implied volatility of 40%. Suddenly the implied volatility drops to 35%, but the stock price does not change. How much money did you just make or lose?”
4. **[2 points]** You have just written call options on 10,000 shares of DGT common stock. (That is, you have sold 100 option contracts, each on 100 shares.) For a single call option, the delta and gamma of the call option are $\Delta = 0.55$ and $\Gamma = 0.04$. DGT common stock is currently trading at \$40 per share.
- a. Suddenly the stock price rises to \$44 per share. How much money did you just make or lose?
- b. If the price of DGT is \$40 per share, how many shares of DGT should you buy or short-sell in order to create a delta-neutral position?
- c. Suppose you buy or short-sell the stock in order to create the delta-neutral position. Is the vega of the resulting delta-neutral position greater than or less than zero? Please *briefly* explain. [Note: *Briefly* means less than or equal to two sentences.]

5. [4 points] **European Digital options** have payoffs as follows: Calls:

$$Payoff = V_T = \begin{cases} A & \text{if } S_T \geq K \\ 0 & \text{if } S_T < K \end{cases}$$

Puts

$$Payoff = V_T = \begin{cases} 0 & \text{if } S_T > K \\ A & \text{if } S_T \leq K \end{cases}$$

with exercise price K , cash amount A and time to maturity T .

- a) By considering the Black-Scholes equation (or otherwise) Determine a formula for the $t = 0$ values of digital **calls** and **puts** on the S&P 500 Index (Be careful to define all parameters) [2 points]
- b) Also, by considering an appropriate number of long calls with strike K , with $t = 0$ value of $C(K)$, and an appropriate number of short calls with strike of $K + \epsilon$, with $t = 0$ value $C(K + \epsilon)$, show that we can consider the $t = 0$ value of a European Digital call, V_0 , as

$$V_0 = -A \frac{\partial C(K)}{\partial K}$$

[2 points]